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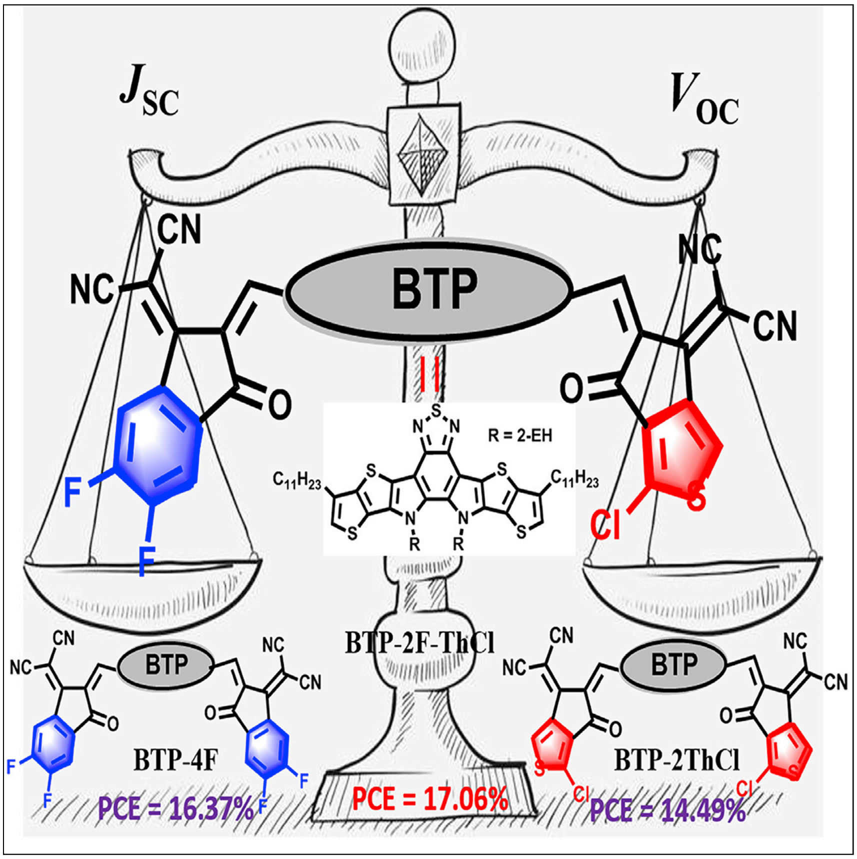
Fine-Tuning Energy Levels via Asymmetric End Groups Enables Polymer Solar Cells with Efficiencies over 17%



Asymmetric acceptor BTP-2F-ThCl-based devices gave the best PCE of 17.06% due to the optimal energy levels relative to those of the devices based on their symmetrical counterparts, BTP-4F (16.37%) and BTP-2ThCl (14.49%).

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HIGHLIGHTS

Two asymmetric SMAs with two distinct end-capping groups were synthesized

The asymmetric BTP-2F-ThCl-based device gave the best PCE of 17.06%

Our strategy can be useful and potentially applied to other material systems

Luo et al., Joule 4, 1236–1247
June 17, 2020 © 2020 Published by Elsevier Inc.
https://doi.org/10.1016/j.joule.2020.03.023

Article

Fine-Tuning Energy Levels via Asymmetric End Groups Enables Polymer Solar Cells with Efficiencies over 17%

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SUMMARY

Generally, it is important to fine-tune the energy levels of donor and acceptor materials in the field of polymer solar cell (PSCs) to achieve a minimal highest occupied molecular orbital (HOMO) energy offset, which yet is still sufficient for charge separation. Based on the high-performance small-molecule acceptor (SMA) of BTP-4F, we modified the end groups of BTP-4F from IC-2F to CPTCN-Cl. It was found that when both end groups were substituted by CPTCN-Cl, the energy level upshift was too large that caused unfavorable energetic alignment, thus poor device performance. By using the strategy of asymmetric end groups, we were able to achieve near optimal energy level match, resulting in higher open-circuit voltage (V_{OC}) and power conversion efficiency (PCE) compared with those given by the PM6:BTP-4F system. Our strategy can be useful and potentially applied to other material systems for maximizing efficiency of non-fullerene PSCs.

INTRODUCTION

Polymer solar cells (PSCs) have seen a rapid revival due to the emergence of high-performance non-fullerene small-molecule acceptors (SMAs) in the last few years.[1–15] The attractive features of SMAs include wide absorption range with absorption edge exceeding 1,000 nm, strong ability to absorb photons, and excellent tunability of chemical structures and morphological properties, which offer great opportunities of promoting open-circuit voltage (V_{OC}), short-circuit current (J_{SC}), and fill factor (FF), simultaneously.[16–29] Among various non-fullerene SMAs, those of ''acceptor-donor-acceptor (A-D-A)''-type architectures were broadly reported for the potential of strong intramolecular charge transfer, appropriate energetics, and excellent planarity.[30–45]

Typically, A-D-A-type SMAs contain three parts: outstretched side chains, a ladder-type donor core, and electron-withdrawing end-capping groups.[45–56] IT-4F is a representative A-D-A-type SMA, which showed power conversion efficiencies (PCEs) over 15% when paired with T1.[28] In addition, BTP-4F (Y6), a special A-D-A-type SMA (or a A-DAD-A-type SMA), bearing a ladder-type electron-deficient core-based central core, exhibits huge potential in the organic photovoltaic field when blended with PM6.[36,37] To further improve the performance of PM6:BTP-4F systems, one of the feasible approaches is to minimize the highest occupied molecular orbital (HOMO) offset, which can be realized by upshifting the HOMO energy level of BTP-4F. However, if this upshift is too excessive, the energetic alignment

Context & Scale

Non-fullerene small-molecule acceptors (SMAs)-based polymer solar cells (PSCs) have attracted considerable attention due to their great potential to realize high power conversion efficiencies (PCEs). To further improve device performance, finely tuning the energy levels of donor and acceptor materials to achieve a minimal energy offset while ensuring sufficient charge separation is important. Herein, we develop an asymmetric SMA with two distinct end groups, namely BTP-2F-ThCl. The energy levels of BTP-2F-ThCl are between its symmetrical counterparts, BTP-4F and BTP-2ThCl. Consequently, in PM6-based OSCs, BTP-2F-ThCl gave a PCE of 17.06%, which is higher than those of BTP-4F (16.37%) and BTP-2ThCl (14.49%). Our strategy can be useful and potentially applied to other material systems for maximizing the efficiency of non-fullerene PSCs.

between PM6 and BTP-4F could be negative for charge separation. Therefore, it is important to finely tune the energy level of the SMA to achieve the minimizing energy level offset (HOMO) while ensuring sufficient charge separation.[18] Generally, introducing electron-donating substituents into the end groups of SMAs, such as methoxyl and methyl, leads to upshifted lowest unoccupied molecular orbital (LUMO), and thereby an enhanced V_{OC} and PCE of the corresponding PSCs.[30,44,45] In addition, employing asymmetric strategy to raise HOMO and LUMO of SMAs is a feasible way.[50–58] Compared with SMAs with asymmetric side chains or central core, developing a SMA with asymmetric terminal acceptor units can not only avoid the time-consuming synthesis but also provide an intuitive method to investigate the structure-property-performance relationship. Interestingly, Tobin J. Marks's group developed two asymmetric acceptors (ITIC-2F and ITIC-3F) with different end groups and compared two SMAs end-group redistribution process in the blend solutions, which emphasizes the importance of understanding the solution-phase stability of ITIC-like acceptors.[58]

Recently, our group developed a chlorinated SMA (ITC-2Cl) with chlorinated thiophene-fused end groups (CPTCN-Cl),[24] from the parent IT-4F incorporating fluorinated benzene-fused end groups. The superior performance of ITC-2Cl-based device is mainly due to its higher V_{OC} (similar J_{SC} and FF). Afterward, we replaced the end groups of BTP-4F with CPTCN-Cl to develop a new SMA (BTP-2ThCl). Unfortunately, BTP-2ThCl-based device exhibited a higher V_{OC}, yet much smaller J_{SC} and FF than the BTP-4F-based device, which may be the result of insufficient charge separation.[59]

Based on the above consideration, to minimize energy level offset (HOMO) while ensuring sufficient charge separation, we develop a new SMA, namely BTP-2F-ThCl, with two distinct end-capping groups (IC-2F and CPTCN-Cl) based on BTP-4F and BTP-2ThCl (Scheme 1). The optical, electrochemical, and crystallization characteristics of BTP-2F-ThCl are closer to BTP-4F, relative to BTP-2ThCl. Consequently, photovoltaic data of the BTP-4F-, BTP-2F-ThCl-, and BTP-2ThCl-based devices have been collected and compared. BTP-2F-ThCl-based device gave a best performance with a PCE as high as 17.06%, along with a V_{OC} of 0.869 V, a J_{SC} of 25.38 mA/cm^2, and an *FF* of 0.774. Meanwhile, the PCEs of BTP-4F- and BTP-2ThCl-based devices were 16.37% and 14.49%, respectively. The higher PCE for the BTP-2F-ThCl-based device is mainly ascribed to the larger value of $V_{OC} \times J_{SC}$ (22.06) than those of the devices based on BTP-4F (21.18) and BTP-2ThCl (20.76). In addition, we also applied this strategy to the traditional A-D-A-type SMA and thus developed an asymmetric SMA of IT-2FCl, which has the same central core as IT-4F and ITC-2Cl. Similarly, IT-2FCl-based device displays a better PCE than IT-4F- and ITC-2Cl-based devices because of the more balanced V_{OC} and J_{SC}. This work indicates that incorporating two distinct end groups into SMAs is conducive to achieving a balance between V_{OC} and J_{SC}, thus leading to a better PCE.

RESULTS AND DISCUSSION

As depicted in Scheme S1, BTP-2F-ThCl is obtained via Knoevenagel condensation reactions between the di-aldehydes BTP-4F-CHO and two different end-capping groups. IT-2FCl was fully characterized by ^{1}H NMR, ^{13}C NMR, and MALDI-TOF (Figures S1–S6). BF-2FCl shows good solubility in common solvents, such as toluene, dichloromethane (DCM), chloroform (CF), and chlorobenzene (CB).

The absorption spectra of BTP-4F, BTP-2F-ThCl, and BTP-2ThCl in dilute CF solution and in film are shown in Figure 1, and the corresponding data are summarized in

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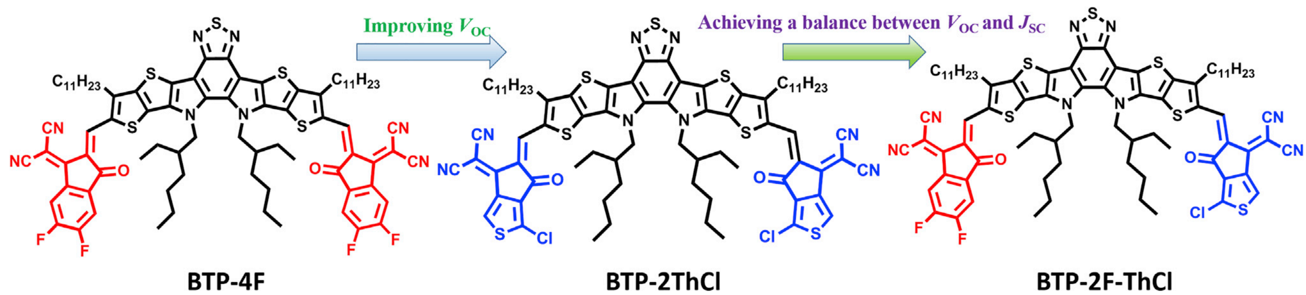
https://doi.org/10.1016/j.joule.2020.03.023



Scheme 1. Chemical Structures of BTP-4F, BTP-2ThCl, and BTP-2F-ThCl

Table 1. As exhibited in Figure 1A, IT-2FCl in dilute CF solution (10^{-5} M) displays a maximum extinction coefficient (ε_{max}) of 2.54×10^5 M^{-1} cm^{-1} at 741 nm, slightly larger than BTP-4F (2.39×10^5 M^{-1} cm^{-1} at 733 nm) and BTP-2ThCl (2.28×10^5 M^{-1} cm^{-1} at 733 nm). In comparison with the solution absorption, the film absorption of BTP-4F, BTP-2F-ThCl, and BTP-2ThCl exhibited an obvious redshift due to the intermolecular π–π packing. As a result, the absorption edge of BTP-2F-ThCl (926 nm) neat film is between those of BTP-4F (930 nm) and BTP-2ThCl (918 nm) (Figure 1B). PM6 (Figure 1C), a high-performance wide-band-gap polymer donor, with a strong absorption of approximately 350–650 nm, was chosen to pair with these three SMAs.

The electrochemical energy levels of BTP-4F, BTP-2F-ThCl, and BTP-2ThCl were investigated via electrochemical cyclic voltammetry (CV) measurements (Figure 1D). The onset oxidation and reduction potentials of BTP-4F, BTP-2F-ThCl, and BTP-2ThCl versus Ag/

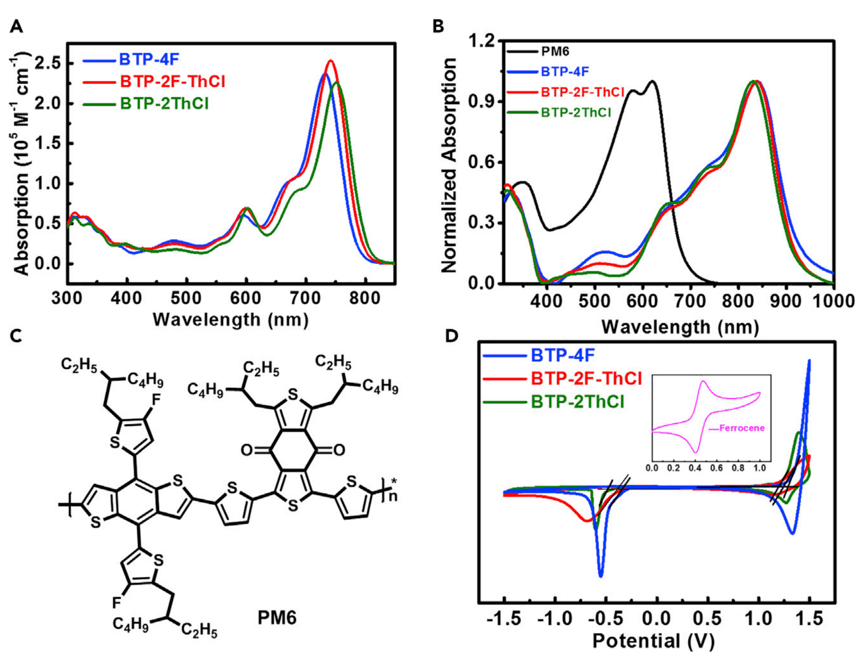


Figure 1. Molecular Structure of PM6 and Photophysical Properties of BTP-4F, BTP-2F-ThCl, BTP-2ThCl, and PM6

(A) Absorption spectra of BTP-4F, BTP-2F-ThCl, and BTP-2ThCl in $CHCl_3$ solution.

(B) Normalized absorption spectra of BTP-4F, BTP-2F-ThCl, BTP-2ThCl, and PM6 in film.

(C) Chemical structure of PM6.

(D) CV curves of BTP-4F, BTP-2F-ThCl, and BTP-2ThCl.

Table 1. Optical and Electrochemical Properties of BTP-4F, BTP-2F-ThCl, and BTP-2ThCl

Acceptor	Λ_{max}^{sol} (nm)	λ_{max}^{film} (nm)	λ_{onset}^{film} (nm)	E_g^{opt}[a] (eV)	$HOMO^{DFT}$ (eV)	$LUMO^{DFT}$ (eV)	$HOMO^{CV}$ (eV)	$LUMO^{CV}$ (eV)	μ_e[b] (10^{-4} cm^2 V^{-1} s^{-1})
BTP-4F	733	841	930	1.33	−5.85	−3.82	−5.72	−4.01	6.06 ± 0.21
BTP-2F-ThCl	741	839	926	1.34	−5.83	−3.81	−5.70	−3.99	5.56 ± 0.25
BTP-2ThCl	750	829	918	1.35	−5.80	−3.80	−5.67	−3.95	4.84 ± 0.31

[a]Calculated from E_g^{opt} =1,240/λ_{onset}.
[b]The average values were obtained from 10 devices.

Ag^+ were found to be 1.36/−0.35, 1.34/−0.37, and 1.31/−0.41 eV, respectively. The measured relative energy level of Fc/Fc^+ and assumed vacuum energy level were 0.44 and −4.8 eV, respectively. Therefore, the corresponding HOMO and LUMO levels of BTP-4F, BTP-2F-ThCl, and BTP-2ThCl are calculated to be −5.73 and −4.01, −5.70 and −3.99, and −5.67 and −3.95 eV, respectively. Clearly, both the HOMO and LUMO energy levels showed an upward trend from BTP-4F, BTP-2F-ThCl to BTP-2ThCl, which is consistent with the previous reported results. Density functional theory (DFT) at the B3LYP/6–31G** level was carried out to investigate the molecular orbital levels of BTP-4F, BTP-2F-ThCl, and BTP-2ThCl, and the long alkyl chains were simplified to methyl units (Figure S7). The calculated HOMO and LUMO energy levels were −5.85 and −3.82 eV for BTP-4F, −5.83 and −3.81 eV for BTP-2F-ThCl, and −5.80 and −3.80 eV for BTP-2ThCl, respectively, in line with the results from CV measurements. The HOMOs and LUMOs of the three acceptors are mainly distributed over the whole molecules, which effectively facilitates charge transport.

To explore the crystallinity and molecular stacking of BTP-4F, BTP-2F-ThCl, and BTP-2ThCl, grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements were performed on their neat films. As shown in Figure 2, the neat films of BTP-4F, BTP-2F-ThCl, and BTP-2ThCl show a preferential face-on π-π stacking, which facilitates electron transport. Compared with BTP-4F, BTP-2F-ThCl displays very similar lamellar stacking peak at $q \approx 0.28$ Å^{-1} in the in-plane (IP) direction and π-π stacking peak at $q \approx 1.75$ Å^{-1} in the out-of-plane (OOP) direction but slightly

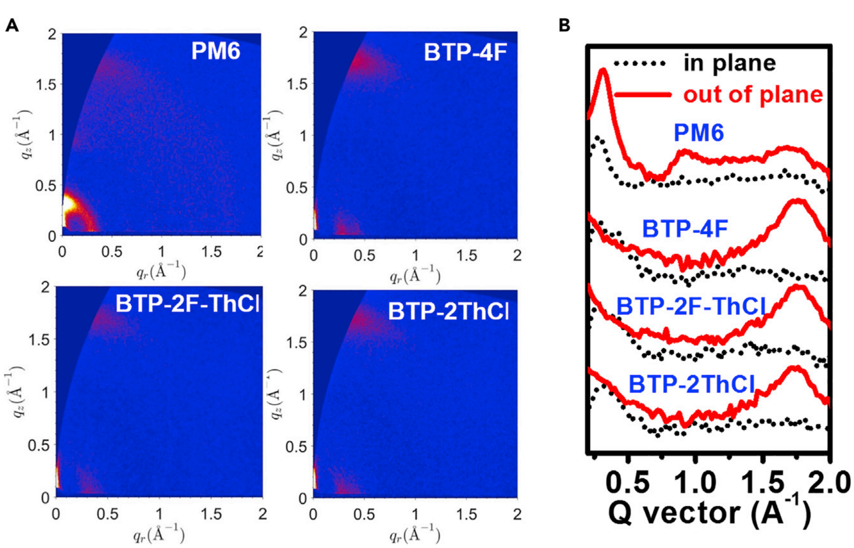


Figure 2. The GIWAXS Characterization of the Optimized Blend Films
(A) GIWAXS patterns of PM6, BTP-4F, BTP-2F-ThCl, and BTP-2ThCl neat films.
(B) Corresponding intensity profiles of PM6, BTP-4F, BTP-2F-ThCl, and BTP-2ThCl neat films along the in-plane (in black) and out-of-plane (in red) directions.

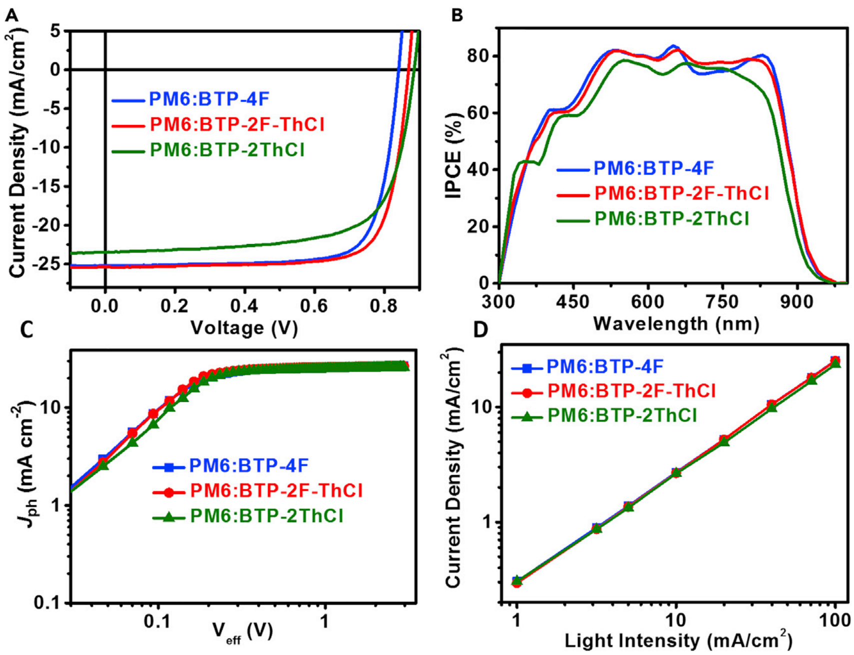


Figure 3. The Photovoltaic Properties of the PM6:BTP-4F-, PM6:BTP-2F-ThCl-, and PM6:BTP-2ThCl-Based OSCs

(A) *J-V* characteristics of the best OSCs under the illumination of AM 1.5G, 100 mW cm^{-2}.

(B) IPCE spectra of the best devices.

(C) J_{ph} versus V_{eff} of the optimized devices.

(D) Light intensity dependence of J_{SC} of the devices.

smaller crystal coherence length (CCL, according to Scherrer's equation)[60] of lamellar peaks (16.1 Å for BTP-4F and 15.7 Å for BTP-2F-ThCl) and π-π stacking (43.5 Å for BTP-4F and 40.3 Å for BTP-2F-ThCl), indicating the slightly weaker crystallinity of BTP-2F-ThCl. The scattering intensity and area of the OOP π-π peaks suggest the weakest crystallinity of BTP-2ThCl among the three acceptors. Consequently, the space-charge-limited current (SCLC) method was utilized to survey the electron mobilities (μ_e) of BTP-4F, BTP-2F-ThCl, and BTP-2ThCl (Figure S8), and corresponding data are summarized in Table 1. The electron mobilities of BTP-4F, BTP-2F-ThCl, and BTP-2ThCl were calculated to be (6.06 $\pm$ 0.21) × 10^{-4}, (5.56 $\pm$ 0.25) × 10^{-4}, and (4.84 $\pm$ 0.31) × 10^{-4} cm^2 V^{-1} s^{-1} from 10 devices, respectively, in accordance with the gradually reduced molecular packing. These results demonstrate that the asymmetric acceptor, BTP-2F-ThCl, shows moderate electron mobility, but it is more biased toward BTP-4F instead of BTP-2ThCl.

To assess the photovoltaic properties of BTP-4F, BTP-2F-ThCl, and BTP-2ThCl, a series of binary PSCs were fabricated with a conventional device structure of indium tin oxide (ITO), (ITO)/poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS)/PM6:acceptor/ PNDIT-F3N/Ag. The optimal active layers were obtained by casting the hot blend solution of PM6:BTP-4F (or BTP-2F-ThCl or BTP-2ThCl) (1:1.2, w/w) with a total concentration of 15 mg mL^{-1} in chloroform, and 0.5% (vol %) 1-chloronaphthalene was utilized as the solvent additive, followed by thermal annealing at 90°C for 5 min. The optimal *J-V* curves of PSCs are exhibited in Figure 3A, and the related data are outlined in Table 2.

Basically, PM6:BTP-4F-based device delivers a PCE of 16.37%, with a V_{OC} of 0.841 V, a J_{SC} of 25.19 mA/cm^2 along with a FF of 0.773. Compared with PM6:BTP-4F-based

Table 2. The Optimized Photovoltaic Performances of PSCs Based on PM6: Acceptors under Standard AM 1.5G Illumination, 100 mWcm^{-2}

Devices	V_{OC} (V)	J_{SC} (mA cm^{-2})	FF	PCE_{max} (PCE_{avg}) %
PM6/BTP-4F	0.841 (0.837 ± 0.002)	25.19 (25.40 ± 0.15)	0.773 (0.761 ± 0.007)	16.37 (16.18 ± 0.17)
PM6/BTP-2F-ThCl	0.869 (0.869 ± 0.005)	25.38 (25.37 ± 0.28)	0.774 (0.761 ± 0.008)	17.06 (16.77 ± 0.19)
PM6/BTP-2ThCl	0.885 (0.884 ± 0.003)	23.46 (23.38 ± 0.35)	0.698 (0.693 ± 0.006)	14.49 (14.33 ± 0.18)

The average values and standard deviations were obtained from 20 devices.

device, BTP-2F-ThCl-based device yields a higher V_{OC} of 0.869 V, similar J_{SC} of 25.38 mA/cm^2 and FF of 0.774, and thus results in a better PCE of 17.06%, which is the highest value for the asymmetric non-fullerene SMAs-based PSCs. The champion PCE was further confirmed by two different research groups to reproduce the results, namely, Min Jie's group and Zhang Jian's group (Figure S9). As for PM6:BTP-2ThCl-based device, it affords a much lower PCE of 14.49% than those of BTP-4F- and BTP-2F-ThCl-based devices, which is mainly ascribed to its inferior J_{SC} and FF. Apart from the device performance, the shelf-life and photostability of PM6:BTP-4F, PM6:BTP-2F-ThCl, and PM6:BTP-2ThCl-based PSCs were investigated (Figure S10). The packaged BTP-2ThC-based device in air can maintain 74% PCE after 30 days, which is slightly higher than those of BTP-2F-ThCl- (72%) and BTP-4F-based devices (68%). The packaged BTP-2ThC-based device under LED illumination can maintain 86.4% PCE after 108 h, similar to BTP-2F-ThCl- (85.6%) and BTP-4F-based devices (84.5%). These results indicate that differences of end groups in Y6-like SMAs have less effect on device stability.

As displayed in Figure 3B, the incident photon-to-current efficiency (IPCE) measurements were carried out to verify the accuracy of J_{SC} values from the *J-V* curves. BTP-4F- and BTP-2F-ThCl-based devices show strong photoresponse in the range of 400–900 nm, obviously higher than those of BTP-2ThCl-based devices. The integrated J_{SC} values from IPCE spectra of BTP-4F-, BTP-2F-ThCl-, and BTP-2ThCl-based PSCs are calculated to be 25.03, 24.97, and 22.93 mA cm^{-2}, respectively, which match well with the J_{SC} values obtained from the *J–V* experiments (within 3.0% mismatch).

To gain insight into the charge collection and exciton dissociation of these three binary systems, the relationship between effective voltage (V_{eff}) and photocurrent density (J_{ph}) was investigated (Figure 3C), and corresponding parameters are summarized in Table S1, including the saturated current (J_{sat}), charge collection efficiency (η_{coll}) and exciton dissociation efficiency (η_{diss}). BTP-2F-ThCl-based device yields a similar J_{ph} of 26.70 mA/cm^2 as BTP-4F-based device (26.60 mA/cm^2), higher than their counterparts using BTP-2ThCl (25.80 mA/cm^2). The η_{coll} and η_{diss} were calculated according to the equation ($\eta = J_{ph}/J_{sat}$) under the maximum power output and short-circuit conditions, respectively. The η_{coll}/η_{diss} values are calculated to be 83.9%/94.7%, 86.3%/95.0%, and 73.3%/90.0% for BTP-4F-, BTP-2F-ThCl-, and BTP-2ThCl-based devices, respectively. The similarity of η_{diss} and η_{coll} for the PM6:BTP-4F- and PM6:BTP-2F-ThCl-based devices could account for their similar FF. The bimolecular recombination in BTP-4F-, BTP-2F-ThCl-, and BTP-2ThCl-based devices was investigated by measuring *J-V* characteristics of the three devices under different light intensities (*P*). The relationship between J_{SC} and *P* follows a power-law formula: $J_{SC} \propto P^S$. As shown in Figure 3D, the exponent S values of BTP-4F-, BTP-2F-ThCl-, and BTP-2ThCl-based devices are 0.965, 0.971, and 0.946, respectively, which clearly indicate the weaker bimolecular recombination of the former two and accord with the higher FF in BTP-4F- and BTP-2F-ThCl-based devices.

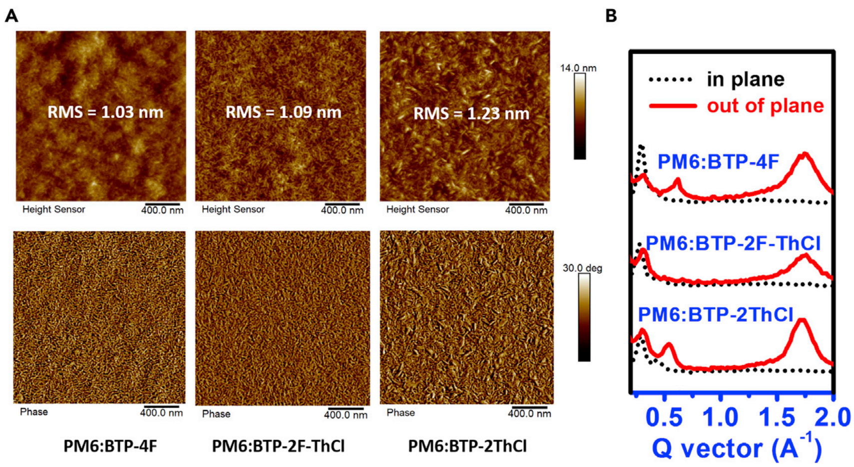


Figure 4. Blend Film Morphologies

(A) AFM height images (top) and AFM phase images (bottom).

(B) GIWAXS intensity profiles along the in-plane (black lines) and out-of-plane (red lines) directions.

The SCLC method was employed to investigate charge transport of blend films based on PM6:BTP-4F, PM6:BTP-2F-ThCl, and PM6:BTP-2ThCl (Figure S11; Table S2). The hole (μ_h) and electron (μ_e) mobilities of the three blend films show the same order of magnitude. From BTP-4F-, BTP-2F-ThCl- to BTP-2ThCl-based blend, the hole and electron mobilities gradually and slightly descend. Additionally, BTP-4F- and BTP-2F-ThCl-based blends yield more balanced mobilities (μ_h/μ_e = 1.60 for BTP-4F; μ_h/μ_e = 1.58 for BTP-2F-ThCl) than BTP-2ThCl-based blend (μ_h/μ_e = 1.87), which contribute to the higher FF in BTP-4F-and BTP-2F-ThCl-based devices.

In addition to charge separation, recombination, and transport, bulk-heterojunction (BHJ) morphology also plays an important role in determining the performance of PSCs. Atomic force microscopy (AFM) in tapping mode was employed to investigate the active-layer morphology (Figure 4A). As shown in AFM height images, PM6:BTP-4F, PM6:BTP-2F-ThCl, and PM6:BTP-2ThCl blend films have uniform and smooth surfaces, and the PM6:BTP-2F-ThCl blend shows a similar root-mean-square (RMS) roughness of (1.09 nm) with PM6:BTP-4F blend (1.03 nm), smaller than BTP-2ThCl-based blend (1.23 nm). PM6:BTP-4F- and PM6:BTP-2F-ThCl-based blends exhibit similar nanofibrillar structures in AFM phase images; by contrast, the PM6:BTP-2ThCl-based blend yields an overly obvious nanofiber structure, which is not conducive to the charge transport.

To investigate the crystallinity and molecular packing of the active layer, GIWAXS measurements were hereby employed (Figures 4B and S12). The PM6:BTP-4F-, PM6:BTP-2F-ThCl-, and PM6:BTP-2ThCl-based blends display a typical face-on orientation, similar to their pure films that we discussed above. In the OOP direction, BTP-2F-ThCl-based blend shows a similar π-π stacking peak at 1.72 Å^{-1} with BTP-4F-based blend but a smaller CCL (51.5 Å for PM6:BTP-4Fand 43.5 Å for PM6:BTP-2F-ThCl), which is in accordance with the results of acceptor pristine films. Additionally, the BTP-2F-ThCl-based blend exhibits a smaller CCL relative than the PM6:BTTT-2FCl-based blend. As for the PM6:BTP-2ThCl blend, the PM6 diffraction signals were quite close to BTP-2ThCl in the OOP direction, and therefore it is hard to clearly separate them out in the PM6:BTP-2ThCl-based blend. However, in the IP direction, the BTP-2ThCl-based blend gives an obvious larger d-spacing than the BTP-4F- and BTP-2F-ThCl-based blends, which indicates the weaker crystallinity of

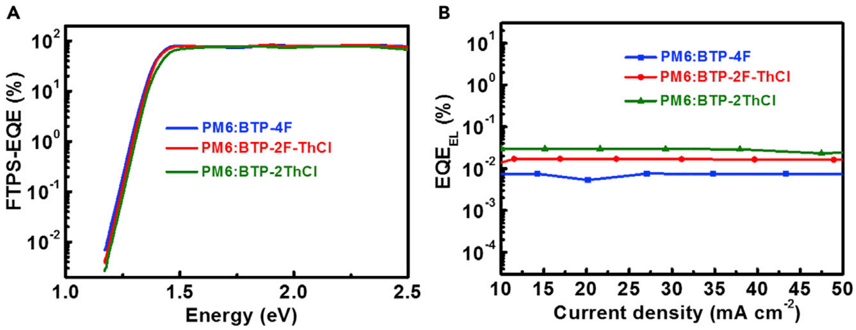


Figure 5. The Experiments of Energy Loss

(A) FTPS-EQE spectra.

(B) EL quantum efficiencies of the cells at various injection current densities.

BTP-2ThCl and accords with the information of the pure acceptor film. These results indicate that PM6:BTP-2F-ThCl blend film shows similar molecular packing and crystallite quality to the BTP-4F-based blend, which can be supported by AFM, SCLC, and GIWAXS tests and leads to the similar FF of the corresponding devices.

Considering the miscibility between the donor and the acceptor has a great effect on the device performance, we performed contact angle measurements to investigate the miscibility and interaction among different materials. As shown in Figure S13 and Table S3, the surface free energy (γ) calculated from the contact angles on water (H_2O) and ethylene glycol (EG) are 27.4, 33.1, 21.8, and 31.4 mN m^{-1} for PM6, BTP-4F, BTP-2F-ThCl, and BTP-2ThCl, respectively. Additionally, according to the Flory-Huggins model, the interaction parameters (χ)[61] are calculated to be 0.27 for $\chi_{PM6,\ BTP\text{-}4F}$, 0.31 for $\chi_{PM6,\ BTP\text{-}2F\text{-}ThCl}$, and 0.13 for $\chi_{PM6,\ BTP\text{-}2ThCl}$, respectively. These results indicate that PM6 shows a stronger miscibility between BTP-2ThCl relative to BTP-4F and BTP-2F-ThCl, which may not result in appropriate domain size, and thereby suffer severe charge recombination. To confirm the difference of domain size, the grazing-incidence small-angle X-ray scattering (GISAXS) experiments were carried out (Figure S14).[62] The pure phase domain sizes (2Rg) for the PM6:BTP-4F, PM6:BTP-2F-ThCl, and PM6:BTP-2ThCl blends are estimated to be 23.2, 23.2, and 14.1 nm, respectively. The small pure phase domain size for PM6:BTP-2ThCl blend may be harmful for the charge separation, and thereby results in severe charge recombination. Additionally, the intermixing domain sizes are calculated to be 60.2, 51.3, and 32.6 nm for PM6:BTP-4F, PM6:BTP-2F-ThCl, and PM6:BTP-2ThCl blends, respectively. It is worth mentioning that PM6:BTP-4F shows similar domain size and interaction parameters compared to that of PM6:BTP-2F-ThCl, which should be a reason for the close FF of the PM6:BTP-4F- and PM6:BTP-2F-ThCl-based devices. These results are in agreement with the data from the contact angle measurements.

To further insight into the reasons for the difference of the V_{OC} in our SMA-PSCs, we performed Fourier transform photocurrent spectroscopy external quantum efficiency (FTPS-EQE) and EL experiments (Figure 5), and the related data are outlined in Table S4. According to previous work,[34,41] the E_{loss} satisfies the following equation:

$$E_{loss} = (E_{gap} - q\Delta VSQ\ oc) + (q\Delta Vrad\ oc) + (q\Delta Vnon\text{-}rad\ oc) = \Delta E_1 + \Delta E_2 + \Delta E_3$$

The first part, ΔE_1, comes from radiative recombination loss above the band gap, which is unavoidable based on the Shockley-Queisser (SQ) theory. In this work, BTP-4F-, BTP-2F-ThCl-, and BTP-2ThCl-based devices exhibit same ΔE_1 (0.26 eV).

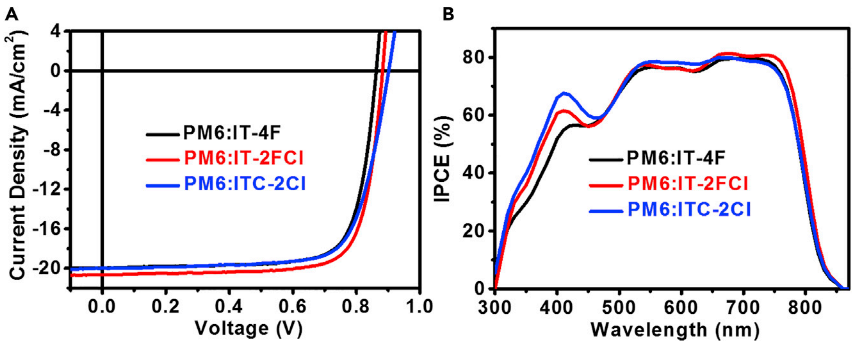


Figure 6. Photovoltaic Characteristics of the IT-4F-, IT-2FCl-, and ITC-2Cl-Based Devices
(A) *J-V* characteristics of the best OSCs under the illumination of AM 1.5G, 100 mW cm^{-2}.
(B) IPCE spectra of the best devices based on PM6:IT-4F, PM6:IT-2FCl and PM6:ITC-2Cl.

The second part, radiative recombination loss below the gap (ΔE_2), depends on the absorption below the band gap, and the three devices show similar ΔE_2 (0.05 eV). The third part of the nonradiative recombination loss (ΔE_3) is obtained via the following equation: $\Delta E_3 = -kT\ln(EQE_{EL})$. To accurately evaluate E_{loss}, we extract the optical band gaps (E_g^{PV}) from the derivatives of the EQE spectra,[41] and E_g^{PV}s of BTP-4F-, BTP-2F-ThCl-, and BTP-2ThCl-based devices are 1.40, 1.40, 1.41 eV, respectively. As shown in Table S4, the BTP-2F-ThCl-based device yields a smaller nonradiative recombination loss of 0.22 eV as compared with that of BTP-4F-based device (0.25 eV), thereby a lower energy loss of 0.52 eV.

Apart from applying this strategy to the BTP-4F system, we have employed this method to the traditional A-D-A-type SMA and thereby developed an asymmetric SMA of IT-2FCl with CPTCN-Cl and IC-2F termini, which is the analog of IT-4F and ITC-2Cl (Scheme S1). In fact, IT-2FCl with IT core will slowly decompose into its symmetric counterpart during the period of purifying by column chromatography on silica gel, whereas BTP-2F-ThCl with BTP core will not decompose; this is why BTP-2F-ThCl has a higher yield (40%) than IT-2FCl (30%). This result also indicates that BTP-2F-ThCl shows slower kinetic of the interconversion relative to IT-2FCl. As shown in Figure 6, IT-2FCl-based device yields a PCE of 14.01%, with a V_{OC} of 0.882 V, a J_{SC} of 20.69 mA/cm^2 along with a FF of 0.768, exceeding devices based on PM6:IT-4F (13.18%) and PM6:ITC-2Cl (13.27%). It is worth mentioning that the PM6:IT-2FCl shows a larger value of $V_{OC} \times J_{SC}$ than IT-4F- and ITC-2Cl-based devices, which is an important reason for the highest PCE in IT-2FCl-based device. The integrated J_{SC} values of IT-4F-, IT-2FCl-, and ITC-2Cl-based devices are 19.40, 20.06, and 19.72 mA cm^{-2}, respectively, matching well with the J_{SC} values obtained from the *J–V* curves (within 2.0% mismatch).

In this work, we develop a new SMA with two distinct end-capping groups based on BTP-4F and BTP-2ThCl, namely BTP-2F-ThCl. BTP-2F-ThCl exhibits similar properties with BTP-4F in optics, electrochemistry, and crystallinity. As a result, BTP-2F-ThCl-based device gave a best performance with a PCE as high as 17.06%, along with a V_{OC} of 0.869 V, a J_{SC} of 25.38 mA/cm^2, and an *FF* of 77.4%. Meanwhile, the PCEs of BTP-4F- and BTP-2ThCl-based devices are 16.37% and 14.49%, respectively. The highest PCE of the BTP-2F-ThCl-based device can be ascribed to the larger value of $V_{OC} \times J_{SC}$ (22.06) than those of the devices based on BTP-4F (21.18) and BTP-2ThCl (20.76). In addition, we applied this strategy to the traditional A-D-A-typed SMA and thus developed an asymmetric SMA, IT-2FCl, which has the same central core as IT-4F and ITC-2Cl. Similarly, IT-2FCl-based device displays a better PCE than those of IT-4F and ITC-2Cl-based devices because of the more balanced V_{OC} and J_{SC}. This contribution shows that employing asymmetric end-group strategy can be useful and applied to many other material systems for maximizing efficiency of non-fullerene PSCs.

EXPERIMENTAL PROCEDURES

Full experimental procedures are provided in the Supplemental Information.

SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at https://doi.org/10.1016/j.joule.2020.03.023.

ACKNOWLEDGMENTS

The work described in this paper is supported by the National Natural Science Foundation of China (nos. 51763017 and 21602150), the Basic and Applied Basic Research Major Program of Guangdong Province (no. 2019B030302007), the Shen Zhen Technology and Innovation Commission (project number JCYJ20170413173814007 and JCYJ20170818113905024), the Hong Kong Research Grants Council (Research Impact Fund R6021-18, project numbers 16305915, 16322416, 606012, and 16303917), and Hong Kong Innovation and Technology Commission for the support through projects ITC-CNERC14SC01 and ITS/471/18. We thank the Design & Manufacturing Services Facility of Hong Kong University of Science and Technology (HKUST) for the help with the contact angle measurements. F.G. acknowledges the financial support from the Swedish Research Council VR (2016-06146). We thank Dr. Cenqi Yan from the Hong Kong Polytechnic University for helping us polish our paper's language.

AUTHOR CONTRIBUTIONS

Z.L., T.L., and H.Y. conceived the idea. Z.L., C.Y., and H.Y. synthesized the two new small molecular acceptors. Z.L., R.M., T.L., and H.Y. drafted this paper. R.M., T.L., and H.Y. were responsible for exploring optimal device fabrication method. R.S., G.X., J.M., and J.Z. contributed to confirming the device performance. Y.X. and X.L. carried out GIWAXS measurements. Y.C., K.C., and G.C. did the CV, AFM, and UV-vis absorption tests. J. Yu, H.S., and F.G. performed FTPS-EQE and EL spectra. J. Yuan and Y.Z. offered acceptor material of Y6. T.L., X.L., and H.Y. provided academic instruction to this project. All authors commented on the final paper.

DECLARATION OF INTERESTS

The authors declare no competing interests.

Received: December 29, 2019
Revised: February 14, 2020
Accepted: March 29, 2020
Published: May 7, 2020

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